

Ashmit Dutta

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EDUCATION

University of Illinois Urbana-Champaign

Aug 2023 – May 2027

BSc, Computer Engineering; Minors in Math & Physics

- **Coursework:** ECE 484 (Safe Autonomy), ECE 408 (GPU Programming), ECE 411 (Computer Organization), ECE 391 (Computer Systems Engineering), CS 444 (Deep Learning).

EXPERIENCE

Assured and Trustworthy AI Research Lab, UIUC

Urbana, IL

Undergraduate Research Assistant (Prof. Huan Zhang)

Dec 2025 – Present

- Contributing Python and CUDA features to **auto-LiRPA**, the core verification library behind α, β -CROWN (VNN-COMP 2021–2025 winner) used for proving safety of learned controllers and perception models.
- Engineering evaluation pipelines and benchmark datasets for the lab's VNN-COMP 2026 global submission.

John Deere

Champaign, IL

Software Engineer, Part-Time Student

Jan 2025 – Jan 2026

- Modernized exporting platform to production for 500+ internal employees on AWS (React, Spring Boot); designed core API integrating PostgreSQL and DB2 across millions of records of fleet/equipment data.
- Integrated AWS CloudWatch observability (reduced MTTR 40%) and CI/CD pipelines (cut deployment time 50%).

The University of Chicago Data Science Institute

Chicago, IL

Machine Learning Research Intern (Mentors: G. Cerati, D. Grzenda, V. Hewes — Fermilab)

Jun 2024 – Aug 2024

- Built a multi-modal Graph Neural Network library in PyTorch Geometric for the Fermilab Exa.TrkX neutrino event reconstruction project, extending the architecture with new message-passing variants for ablation studies.
- Processed 20,000+ detector events with H5Py on UChicago's Slurm HPC cluster for end-to-end training on real sensor data; presented findings at the DSI summer symposium.

PROJECTS

GPU Accelerated Deep Learning Inference (LeNet-5) | C++, CUDA

Feb 2026 – Apr 2026

- Developed high-performance 2D convolution kernels for CNN inference on CUDA GPUs (relevant to onboard perception on edge robotics platforms); converted convolution to GEMM via im2col with kernel fusion.
- Profiled with Nsight Compute/Systems; used shared-memory tiling, CUDA streams, and Tensor Core acceleration for sub-10 ms inference. **Placed 1/130 in ECE 408 class competition for fastest inference kernel.**

Stanford TreeHacks: Multi-Modal Video Search Engine | Python (Modal, BrightData, FastAPI)

Feb 2026

- **Winner (Modal Inference Track + Bright Data AI) at Stanford TreeHacks 2026;** built a distributed video search engine with a multi-modal embedding pipeline fusing vision, audio, and text for semantic retrieval.

Out-of-Order Superscalar RISC-V Processor | SystemVerilog, Design Compiler, RVFI

Jan 2026 – Apr 2026

- Architected a 2-wide superscalar RV32IM processor in SystemVerilog with Explicit Register Renaming: 48-entry physical register file, 16-entry ROB, and 4 reservation stations (ALU/MUL/DIV/MEM); designed memory subsystem with split LSQ (store-to-load forwarding), 2-slot 256-bit linebuffer, and tournament branch predictor (bimodal + GShare, 8-entry BTB, 4-entry RAS).
- Verified against Spike and RVFI across 6 benchmarks with zero mismatches; synthesized in Design Compiler at 500 MHz. **Placed top-10 in ECE 411 class competition with $3.6\times PD^4$ improvement over staff baseline (0.82 CoreMark IPC, peak 1.06).**

LEADERSHIP, PUBLICATIONS & AWARDS

Publication: *Is Human-Written Data Enough? Teaching Reasoning to LLMs Without RL or Distillation* — AI for Math Workshop @ ICML 2025; with NVIDIA researchers.

Co-Founder & Lead Organizer, Online Physics Olympiad (OPhO): largest student-run international physics competition — built a 22k+ Discord community, 3k+ Facebook following, and 3k peak annual competitors since 2020; raised \$10k+ in sponsorships from Citadel, Jane Street, and Wolfram.

Awards & Leadership: 2nd Place, Bradley Mottier Innovation Challenge, UIUC ISE (\$2,000) — AR glasses for ASL translation. Executive Board Member (Treasurer), IEEE Chapter at UIUC.

TECHNICAL SKILLS

Software: Amazon Web Services (AWS), Docker, FastAPI, Spring Boot, MongoDB, Next.js

Hardware: SystemVerilog, RTL design, computer architecture, CUDA, RVFI, Design Compiler, Vivado, FPGA, STM32

Languages: C/C++, SystemVerilog, Assembly (RISC-V), Python, Java, JavaScript, SQL

Tools: Nsight Compute/Systems, Gprof, QEMU, GDB, Linux/Unix, Git, Docker, AWS, LaTeX